Cases of Clinical and radiographic findings of gingival and periodontal diseases

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Learning outcomes

- ***Demonstrate the clinical features of gingival and periodontal diseases***
- ***Illustrate the different patterns of Bone Destruction in Periodontal Diseases***

Periodontal Health and Gingival Health

- Periodontal health is defined as a ***state free from inflammatory periodontal disease*** that allows an individual to function normally and not suffer any consequences (mental or physical) as a result of past disease

- ***Absence of inflammation*** is a pre-requisite for defining health

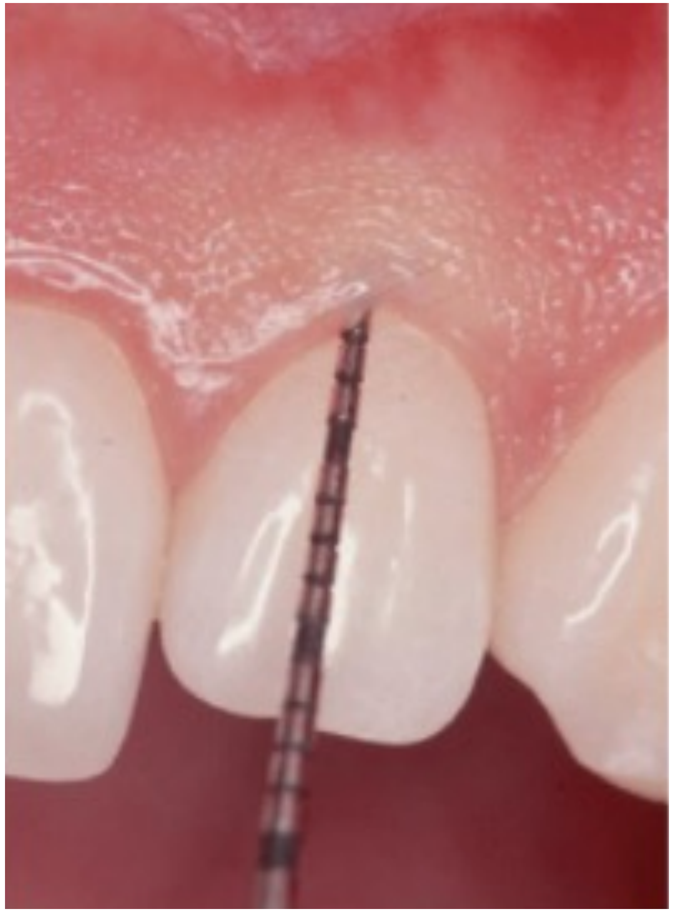
- Periodontal health can exist ***before disease commences*** and, health can be ***restored to an anatomically reduced periodontium***

Determinants of clinical periodontal health

1. *Microbiological* –supra and Subgingival plaque

2. *Host* –Local predisposing factors
--Systemic modifying factors

3. *Environmental* –smoking, medications, stress, nutrition



Periodontal health

- No local factors
- No bleeding on probing
- No changes in color, size, contour, consistency, texture
- Probing depth --- 2-3 mm
- No attachment loss

No radiographic bone loss

Gingival Diseases and Conditions

Gingivitis: Dental Biofilm-Induced	Gingival Diseases: Non-Dental Biofilm-Induced

a. Dental plaque biofilm induced gingivitis

- **Defined as *an inflammatory lesion resulting from interactions between the dental plaque biofilm and the host's immune-inflammatory response, which remains contained within the gingiva and does not extend to the periodontal attachment***

- ***Reversible* – *by reducing levels of dental plaque***

- **Primary parameter for diagnosing gingivitis—*bleeding on probing (BOP)***

Clinical Signs of Plaque-Induced Gingivitis

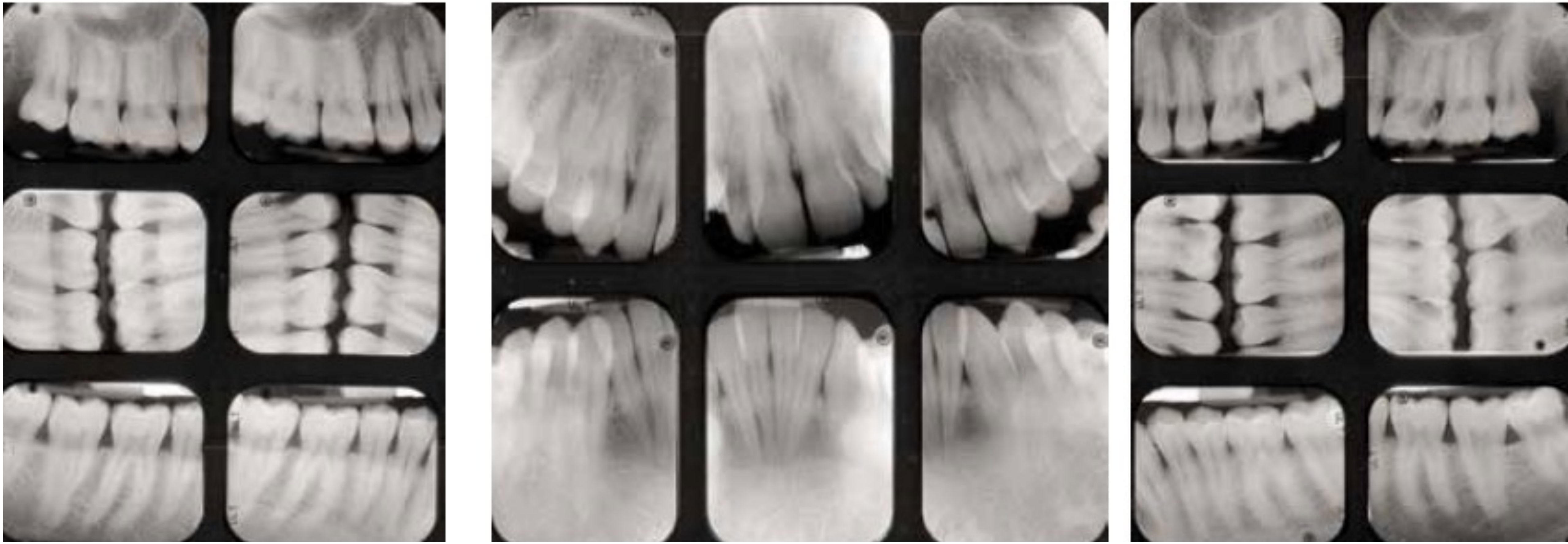
-changes in gingival color, contour, and consistency --redness (erythema), swelling (edema), bleeding, increased gingival crevicular fluid, and tenderness

- Intraoral finding--- dental crowding, changes in the volume and colour of the gum presenting an edematous and reddened gingival margin

- Presence of local factors

- Bleeding on probing

- No attachment loss



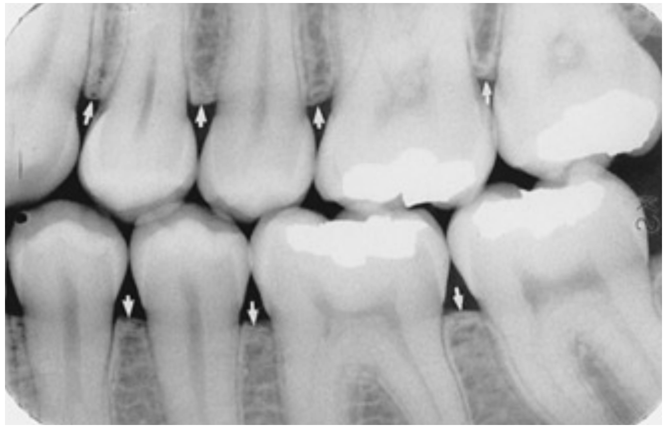
No radiographic bone loss

Gingivitis induced by the dental-plaque biofilm is divided into three categories:

- **Associated with dental biofilm alone**
- **Mediated by systemic or local risk factors**
- **Drug-influenced gingival enlargement.**

Gingivitis induced by the dental-plaque biofilm—Associated with dental biofilm only

- Presence of local factors
- Bleeding on probing
- Changes in color, size, contour, consistency, texture
- Probing depth --- 2-3 mm
- No attachment loss

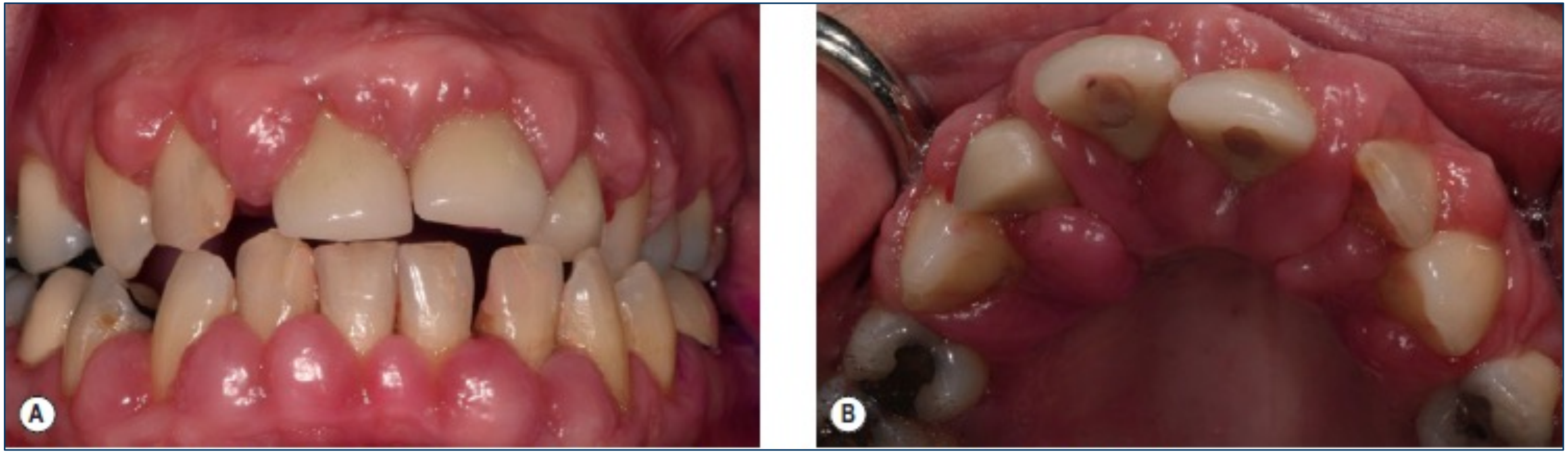


Gingivitis induced by the dental-plaque biofilm--Mediated by systemic or local risk factors

Local risk factors-plaque retentive factors

Systemic risk factors-pregnancy gingivitis

Gingivitis induced by the dental-plaque biofilm—Drug induced gingival enlargement



Three drugs causing gingival enlargement---Phenytoin, Nifedepin, Cyclosporin

Clinical presentation -- predominantly interproximal, nodular, gingival overgrowth covering one-third to one half of the clinical crowns particularly in the anterior teeth

Diagnosis -- drug-induced gingival disease

b. Non-plaque induced gingival lesions

- Non-dental plaque-induced gingival conditions are *not caused by plaque* and usually do not resolve following plaque removal

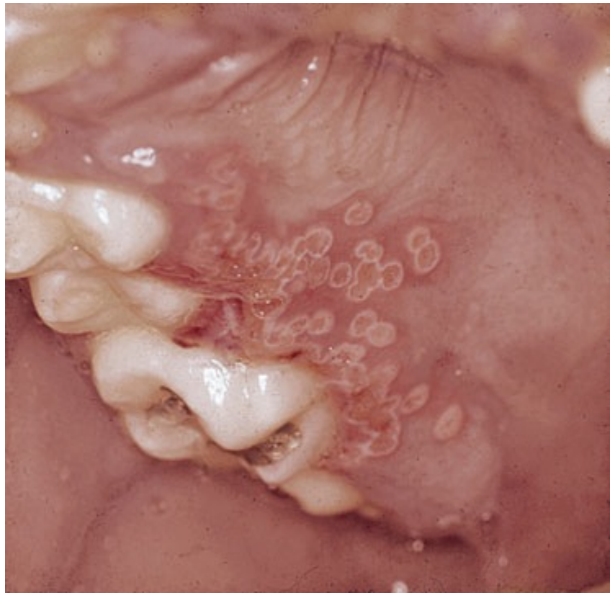
- Lesions may be *manifestations of a systemic condition or may be localized to the oral cavity*

- Although not caused by the dental plaque biofilm, the severity of clinical manifestations often depends on plaque accumulation and subsequent gingival inflammation

- *Less common* but often of major significance for patients

- They represent *pathologic changes limited to gingival tissues*

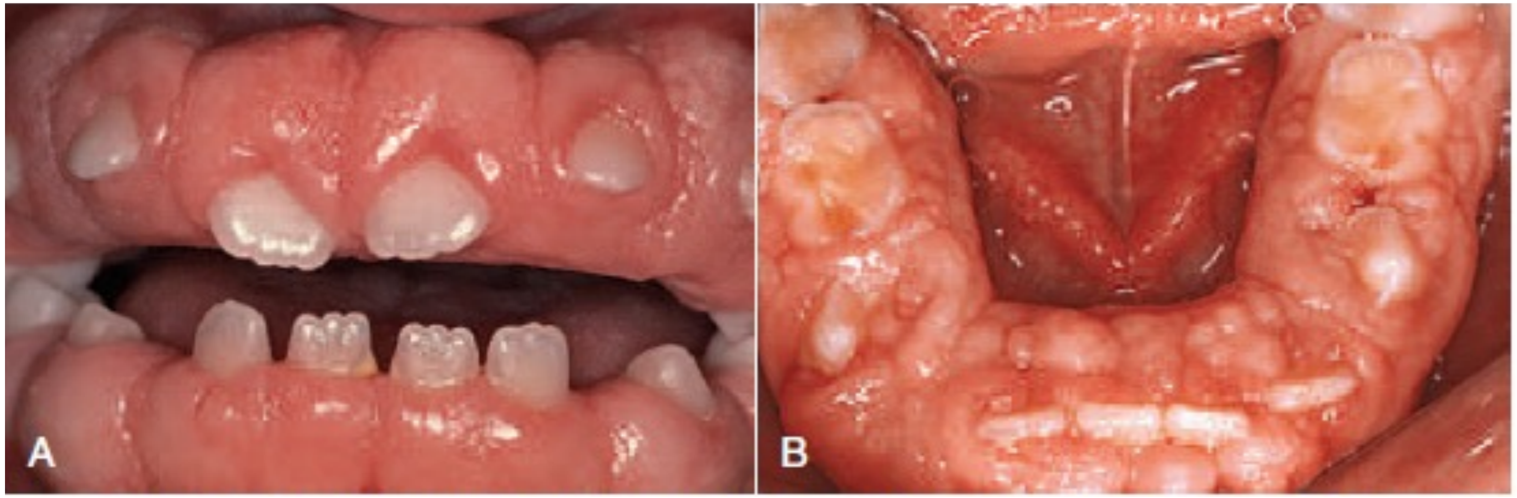
Gingival diseases of specific bacterial origin



Gingival diseases of viral origin

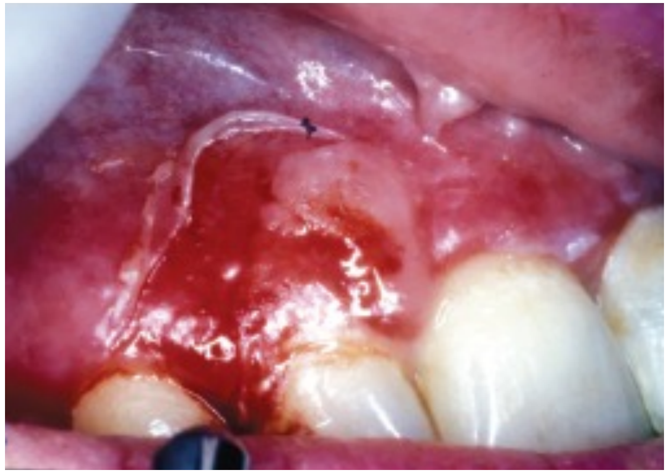
Herpetic lesions develop as intraoral blisters, that quickly burst, leaving minuscule ulcerations

Gingival diseases of fungal origin



Gingival lesions of genetic origin

Inflammatory and immune conditions and lesions



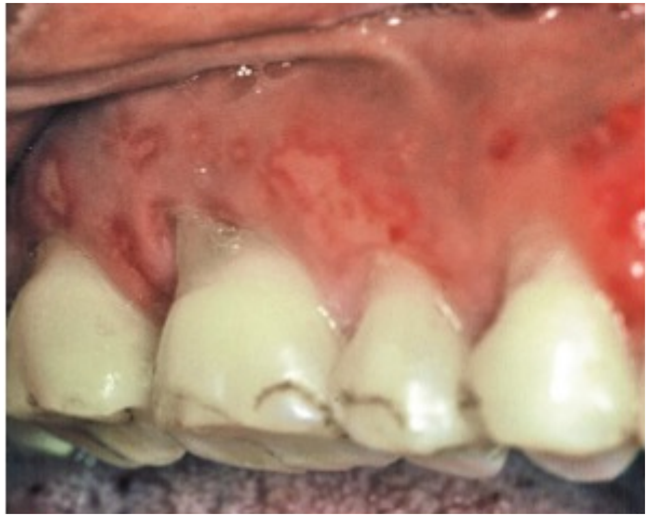
A 62-year-old woman with benign mucous membrane pemphigoid

Contact allergy

Diffuse gingivitis and cheilitis--contact allergy to flavor additive in dentifrice

TRAUMATIC LESION (factitious, iatrogenic, or accidental)

A. Chemical injury
B. Physical injury
C. Thermal injury



Overzealous use of a toothbrush causing painful ulcer

Aspirin burn

Burn

Forms of periodontitis

- **Based on pathophysiology, three different forms:**

  (A) Necrotizing periodontitis

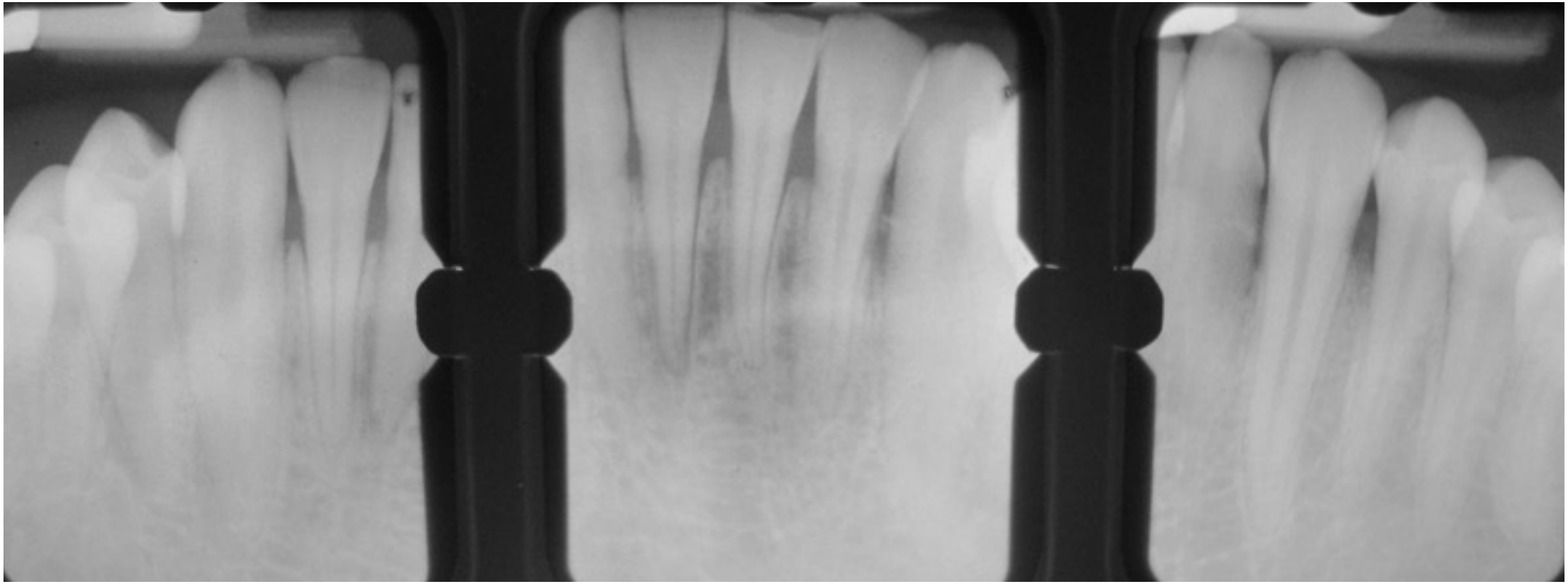
  (B) Periodontitis

  (C) Periodontitis as a direct manifestation of systemic diseases

Necrotizing periodontitis

- A 20-year-old male patient complained of intense and persistent oral pain, gingival bleeding and fetid breath for two weeks
- He was a heavy smoker for 6 years (20 cigarettes per day), and had been experiencing heavy stress due to family problems
- Extraoral examination-- bilateral submandibular lymph nodes tender on palpation and rise in body temperature
- Intraoral examination--poor oral hygiene with plaque, heavy calculus deposits, visible suppuration and severe halitosis

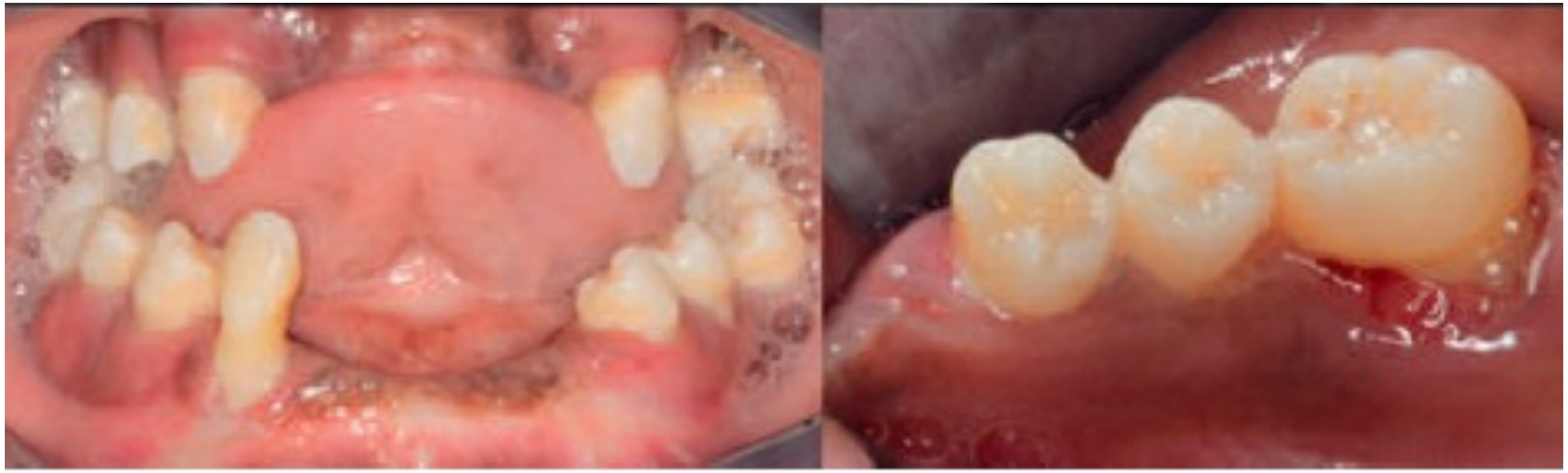
- **Examination of gingiva -- thin whitish film (pseudo-membrane) covering the attached gingiva**
- **Generalized ulcerations and tissue necrosis and formation of characteristic punched out crater like depressions with exposition of the interdental bone**



A periapical radiographic examination showed slight horizontal bone loss in the lower anterior region

Periodontitis as a manifestation of systemic disease

PAPILLON LEFEVRE SYNDROME



PLS is an autosomal recessive disorder
Clinical manifestations --severe periodontitis with hyperkeratosis of palms, soles, or knees
Can affect primary and permanent dentition
An increase in tooth mobility and periodontal abscesses is frequently observed shortly after the eruption of permanent teeth

- Patient with Papillon-Lefèvre (PLS) syndrome exhibiting hyperkeratosis on palms of the hand and soles of the feet. PLS clinically affects both the primary and permanent dentition. Signs of palmoplantar keratoderma usually appear simultaneously with the eruption of the first primary teeth (5–6 months), but may appear as early as 1 month of age.

Panoramic radiograph of a 13-year-old female with Papillon–Lefèvre syndrome (PLS)

Periodontitis

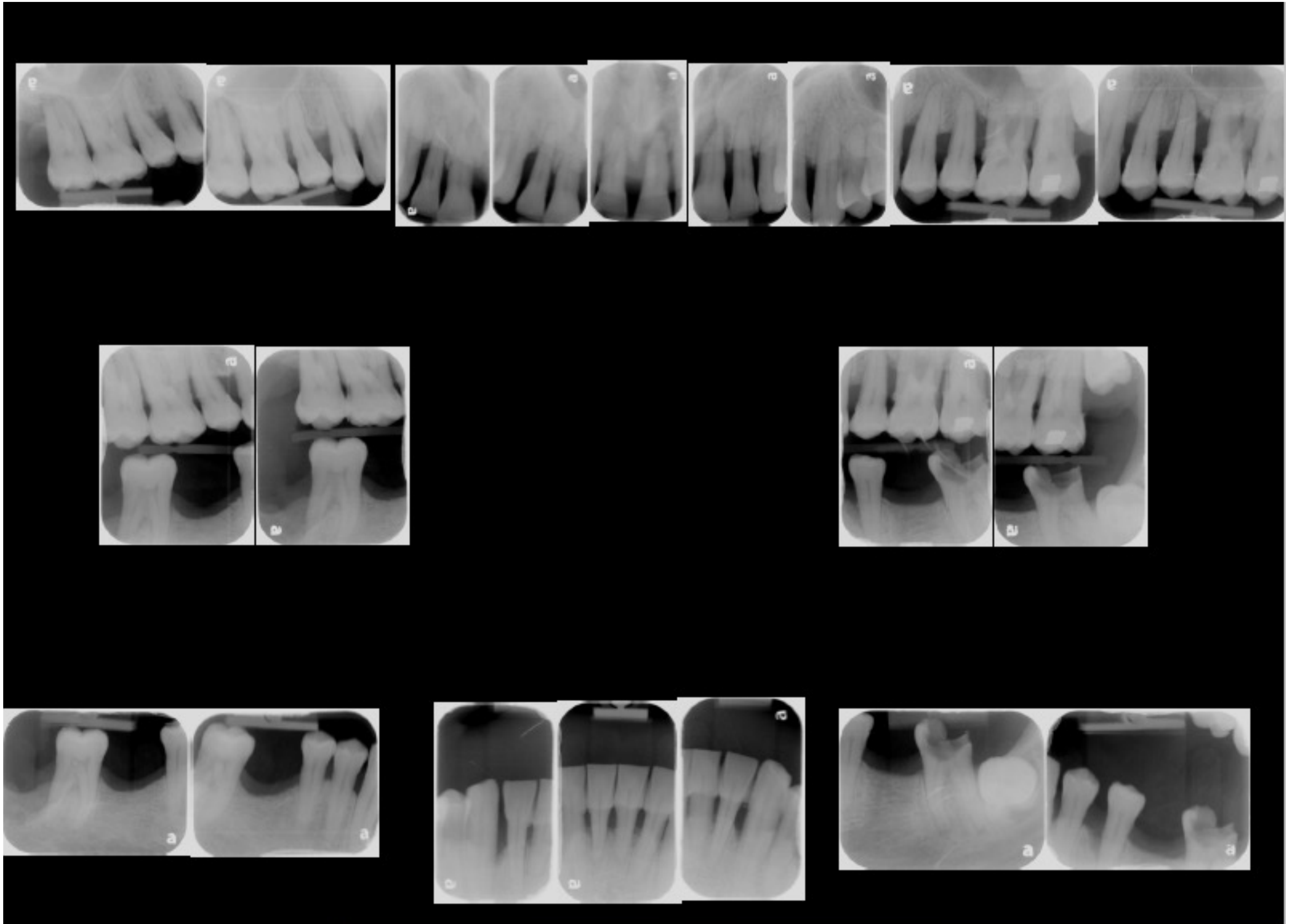
Age --- 58 years Male

Medical history --- not relevant

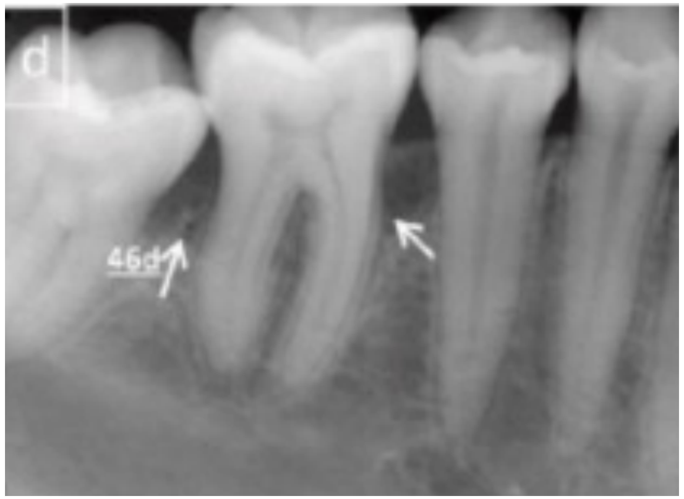
Dental history --- last visit to dentist 10 years back

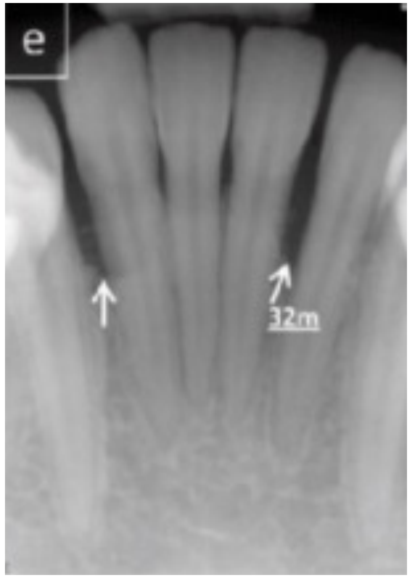
Chief complaint --- many loose teeth, bad breath

Examination – Heavy calculus
Class III occlusion



Molar-incisor pattern periodontitis





Thank you